Adding, Subtracting, and Multiplying Polynomials

Name _____

Algebra 2 Lesson 3-2 (fast track) Date _____ Period _____

Learning goal: I can multiply polynomials and write a polynomial function for a real situation in standard form.

Language goal: I can explain each step using the words Distributive Property, like terms, degree, standard form, and polynomial function.

Spanish cognates: *Propiedad Distributiva, Propiedad Conmutativa, Propiedad Asociativa.*

On the quiz: (1) Bonus: add two polynomials in a and b and write the sum in standard form (IXL 9A3; Example 1 is skipped). (2) Multiply a binomial by a trinomial (Example 2A). (3) Closure of monomials (full Savvas quiz only). (4) Volume of an open box: x times length times width, multiplied out in standard form (Example 4, Example 2B and the Practice). (5) Compare two volume rates (full Savvas quiz only).

On the Topic 3 test: Question 4 asks you to multiply a binomial by a trinomial and write the product in standard form (Example 2). Question 7 asks for the volume function of an open tray folded from a rectangle of paper (Examples 2 and 4).

Example 2: Multiply Polynomials

The idea: use the **Distributive Property**: multiply *every* term of the first polynomial by *every* term of the second. Then group and combine like terms.

A. $(2m+5)(3m^2-4m+2)$

1. Distribute: $2m(3m^2-4m+2)+5(3m^2-4m+2)$
2. Multiply: $6m^3-8m^2+4m+15m^2-20m+10$
3. Group: $6m^3+(-8m^2+15m^2)+(4m-20m)+10$
4. Combine: $\boxed{6m^3+7m^2-16m+10}$

Check with a box: $2\times 3=6$ products.

	$3m^2$	$-4m$	$+2$
$2m$	$6m^3$	$-8m^2$	$4m$
$+5$	$15m^2$	$-20m$	10

Like terms sit on the diagonals:
$-8m^2+15m^2=7m^2$ and $4m-20m=-16m$.

B. $(mn+1)(mn-2)(mn+4)$ Three factors: multiply **two binomials first**, then multiply that product by the third.

1. First two: $(mn+1)(mn-2)=m^2n^2-2mn+mn-2=m^2n^2-mn-2$
2. Distribute the third: $(m^2n^2-mn-2)(mn+4)=m^2n^2(mn+4)+(-mn)(mn+4)+(-2)(mn+4)$
3. Multiply: $m^3n^3+4m^2n^2-m^2n^2-4mn-2mn-8$
4. Group and combine: $m^3n^3+(4m^2n^2-m^2n^2)+(-4mn-2mn)-8=\boxed{m^3n^3+3m^2n^2-6mn-8}$

Degree of a term = sum of its exponents: m^3n^3 (6), $3m^2n^2$ (4), $-6mn$ (2), -8 (0), so the answer is in standard form.

Say it: To multiply, I use the _____ to multiply every term of the first polynomial by every term of the second. Then I combine _____ and write the answer in _____, highest _____ first.

Try It 2: Multiply the polynomials

a. $\left(6n^2 - \frac{1}{2}\right)\left(n^2 + n + \frac{3}{5}\right)$

b. $(mn + 1)(m^2n - 1)(mn^2 + 2)$

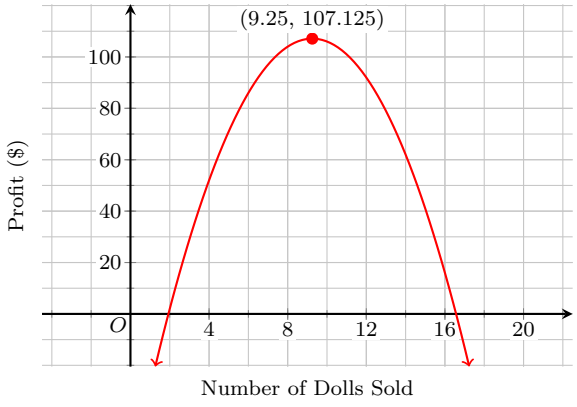
Example 4: Write a Polynomial Function

Hachi makes Seminole Indian dolls to sell at the local street market. As Hachi produces a greater number of dolls, she can lower the cost per unit. The cost c of making x dolls can be represented with the function $c(x) = 12x + 64$. The function $v(x) = 49 - 2x$ relates the price v to the number of dolls produced, x. How many dolls should Hachi sell each week to maximize her profit P? Recall that **profit equals revenue minus cost**.

Formulate. Revenue = price × number sold:
$R(x) = (49 - 2x)x$
Profit: $P(x) = R(x) - c(x) = (49 - 2x)x - (12x + 64)$
Compute. Distribute x, then distribute -1:

1. $P(x) = (49x - 2x^2) - (12x + 64)$
2. $= 49x - 2x^2 - 12x - 64$
3. Combine like terms, standard form:
   $\boxed{P(x) = -2x^2 + 37x - 64}$

Interpret. The domain is the whole numbers: she cannot sell part of a doll, so the vertex (9.25, 107.125) is not the answer. $P(9) = 107$ and $P(10) = 106$.
Sell 9 dolls per week for a weekly profit of $107.



Say it: Profit is revenue minus cost. I use the _____ to distribute x and -1, combine _____, and write the _____ in _____.

Try It 4: A new cost function

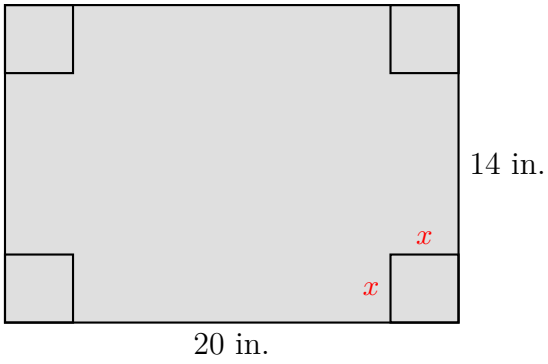
The cost of Hachi's materials changes so that her new cost function is $c(x) = 4x + 42$. Find the new profit function. Then find the quantity that maximizes profit and calculate the profit.

Hint: the vertex of $y = ax^2 + bx + c$ is at $x = -\frac{b}{2a}$; then compare P at the whole numbers on each side, as in Example 4.

Practice: An open tray from sheet metal

A foundry manufactures aluminum trays from pieces of sheet metal as shown. Squares of side x are cut from each corner and the sides are folded up. Let x represent the side length of each square.

a. Write expressions for the length, width, and height of the metal tray.
Length = _____
Width = _____
Height = _____



b. Write and simplify a polynomial function V to represent the volume of the tray.
Volume = length × width × height. Multiply two factors first, as in Example 2B.

$V(x) =$ _____ · _____ · _____

$V(x) =$ _____ x^3 − _____ x^2 + _____ x

IXL: Practice for the quiz

Do the starred skills first. **SmartScore 70 or above on each starred (★) skill replaces your quiz score.**

Code	Skill	Feeds
8GN ★	Multiply polynomials	Quiz 2; Quiz 4 (box volume)
9A3 ★	Add and subtract polynomials	Quiz 1 bonus (add, standard form)
DYB	Polynomial vocabulary	Warm-up: like terms, degree

Extension: Practice 13 (Reason), on your own

Polynomial function R is the difference of two degree-two polynomial functions. What are the possible degrees for R? Explain.
Hint: subtract two degree-two polynomials whose x^2 terms have different coefficients, then two whose x^2 terms have the same coefficient.

The degree of R can be _____. When I subtract, I combine _____. If the x^2 terms have the same coefficient, they cancel, so the _____ of R goes down (when $R \neq 0$; if every term cancels, $R = 0$ has no degree).

Before the quiz, finish these sentences.

To multiply polynomials, I use the _____ and then combine _____.

A polynomial is in standard form when its terms go from highest to lowest _____.

To write a profit _____, I subtract the cost from the revenue and distribute -1 to _____ term of the cost.

The volume of an open box is length $\times$ _____ $\times$ _____; I multiply two factors first and write $V(x)$ in _____.